

COURSE DESCRIPTION			
Program	Polymer Technology		
Course Title	Introduction to Polymer Science		
Course Code	1091	Semester	I
Type of Course	Practicum	Contact Hours	90
Teaching Scheme (L: T:P:R)	3:0:3:0	Credits	4.5
CIE Marks	40	ESE Marks	100

Introduction

Polymers are crucial to modern life, providing lightweight, durable, and versatile materials for everything from clothing and packaging to advanced medical devices, vehicles, and electronics, enhancing comfort, efficiency, and safety while enabling technological progress, though managing their waste through recycling and biodegradable alternatives is a growing focus. This course intends to give students an introduction to different types of polymers, their chemical structure, properties and applications. They also get an opportunity to prepare an album of different types of polymer products available in the market and find out the polymeric material with which the product is made.

Course Objectives

1	To summarise the history, general characteristic and classification of polymers
2	To understand Natural rubber and different forms of natural rubber.
3	To understand about the commercially available Synthetic rubbers
4	To understand about different types of plastics and fibres.

Course Pre-requisite

Topic	Course code	Course Title	Semester
Hydrocarbons, alkane alkene, alkynes			Secondary Education
Polymers			Secondary school

Course Outcome

CO	Course Outcome	Blooms Taxonomy Level
CO1	Explain the history, general characteristic and classification of polymers	Understand
CO2	Explain the structure, property and applications of Natural rubber.	Understand
CO3	Comprehend different types of synthetic rubber, structural formula, properties and applications.	Understand
CO4	Explain the classification, Structural formula, property and applications of Plastics and fibres.	Understand

CO-PO mapping

COURSE ARTICULATION MATRIX											
Course Outcomes	PROGRAM OUTCOMES										
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	-	-	-	-	-	-	-	-	-	3
CO2	3	-	-	-	-	-	-	-	-	-	3
CO3	3	-	-	-	-	-	-	-	-	-	3
CO4	3	-	-	-	-	-	-	-	-	-	3
Total	12	-	-	-	-	-	-	-	-	-	12
Correlation Level	3	-	-	-	-	-	-	-	-	-	3

Legends: 1 - Low; 2 - Medium; 3 - High; No Correlation - "-"

Teaching and Assessment Scheme

Teaching Scheme/Week				Self-learning (SS)/Week	Total Hours/Semester	Credits C	Examination Scheme					
							Theory			Practical		
L	T	P	S				CIE	ESE	Total	CIE	ESE	Total
3	0	3	0	6	90	4.5	20	60	80	20	40	60
							140					

L: Lecture (One unit is of one-hour duration with one credit), T: Tutorial (One unit is of one-hour duration with one credit), P: Practical (One unit is of one-hour duration with 0.5 credit), S: Project (One unit is of one-hour duration with 0.5 credit) SS: Self-Learning, CIE: Continuous Internal Evaluation, ESE: End Semester Examination

Syllabus - Major Topics

Module	Title	Major Topics	Instructional Hours	
			L	P
Module I	History of Polymers and Classification of Polymers.	- History of Polymers - Hydrocarbon - Saturated and unsaturated, aliphatic and aromatic, homologues series, characteristics, Alkanes, Alkenes, Alkynes- their structure - Micromolecules, macromolecules, monomers, oligomers, polymers, repeating units - examples -Polymerisation-Degree of polymerization - molecular weight of polymers - molecular weight of monomers - Relationship between DP and Mw., Simple Problem to find molecular weight of polymers. - Classification of polymers - biological - non biological - Natural, synthetic and derived polymers - Organic and inorganic polymers - homo chain and hetero chain polymers - homo and co - polymers - linear branched and cross linked polymers. Alternating and random co - polymers. Examples of each.	12	0
Module I	History of Polymers and Classification of Polymers.	Specimen preparation of Natural polymers, synthetic polymers, organic polymers, inorganic polymers.	0	12
Module II	Natural Rubber	- Natural rubber-origin-monomer-structure of NR-chemical formula-Properties of Rubber-Molecular weight of NR - Forms of NR-Latex-field latex - concentrated latex, Double centrifuged latex - Various latex industries - Coagulation-preservation of latex - Forms of dry rubber-RSS-TSR-ISNR - Various industries producing marketable forms of NR. - CIS-NR, Trans NR- structure.	11	0
Module II	Natural Rubber	Specimen preparation of RSS, TSR, Demonstrate the Coagulation of NR latex, Demonstrate preservation of NR latex, Prepare 5 samples made of NR with properties and application.	0	11
Module III	Synthetic rubber	Different types of Synthetic rubber available in the market. The structure, raw polymer properties and applications of IR, SBR, BR, IIR, EPDM, NBR, CR, CSM, Silicone rubber.	11	0
Module III	Synthetic rubber	Specimen preparation of products made of SBR, BR., Preparation of samples of products made of IIR, EPDM., Preparation of samples of Product made of NBR, CR., Preparation of samples of Product made of CSM, Silicone.	0	11
Module IV	Plastics and fibres	- Classify plastics -Define Thermoset and thermoplastics with 5 examples each., 5 properties of Thermoplastics and thermosets. - Plastics advantages and disadvantages - Common household articles and plastics used to make them-Kitchen and food storage, bathroom and personal care, cleaning and storage, furniture, Electronic and general goods. - efine fibre - classify fibres.- - Natural and synthetic fibres with 5 examples. - Mention the plastics used to make fibres. and their properties.properties of fibre. Applications of fibre . - what are Liquid resins, adhesives and sealant - Explain with 2 examples. each .	11	0
Module IV	Plastics and fibres	Sample collection of thermoplastics, thermosets, Natural fibres and synthetic fibres.	0	11

Detailed Syllabus

Subtopic	Student Learning Outcome	Mode of delivery (L/T)	Mapped CO	Taxonomy Level	Instructional Hours
Module I					
History of polymers	Describe the history of Polymers	L	CO1	Understand	1
Hydrocarbon	Explain Saturated and unsaturated Hydrocarbons with 3 examples.	L	CO1	Understand	1
Hydrocarbon	Explain aliphatic and aromatic hydrocarbons with 3 examples.	L	CO1	Understand	1
Alkanes, Akenes, Alkynes	Discuss homologous series, alkanes, alkenes and alkynes.	L	CO1	Understand	1

Structure	Learn the structure of alkanes ,alkenes and alkynes.	L	CO1	Understand	1
Macromolecules and micromolecules	Explain what are macromolecules and differentiate them from micromolecules stating their molecular weights .eg. NR and NaCl.	L	CO1	Understand	1
Definition	State the definition of monomers,polymers ,oligomers,repeating units with 2 examples.	L	CO1	Understand	0.5
Polymerisation	Define polymerisation with examples and degree of polymerisation	L	CO1	Remember	0.5
Degree of Polymerisation	State the relationship between Dp and Mw. Do simple problems to find the molecular weight of polymers.	L	CO1	Understand	1
Natural,Synthetic polymer	Classify polymers into Natural ,synthetic and derived polymers with 3 examples.	L	CO1	Understand	0.5
Organic and inorganic	Explain organic and inorganic polymer with example and their Structural formula	L	CO1	Understand	0.5
Types of Polymers	Explain homochain and heterochain polymers with 2 examples and structure.	L	CO1	Understand	1
Homopolymer ,Copolymer	Explain homo and copolymer, random copolymer ,alternating copolymer with 2 examples and structure.	L	CO1	Understand	1
Classification based on structure	Explain Linear ,branched and Crosslinked polymers with one example each.	L	CO1	Understand	1
Natural polymer	Collect 2 samples of natural polymer and mention the name of Polymer, its 3 properties and its application.	P	CO1	Apply	3
Synthetic polymer	Collect 2 samples of synthetic polymer and mention the name of Polymer , its 3 properties and its application.	P	CO1	Apply	3
Organic polymer	Collect 5 samples of organic polymer - PE,PP, Nylon,Polyester,PVC .	P	CO1	Apply	3
Inorganic	Collect samples of Inorganic polymer.- silicone rubber.	P	CO1	Apply	3
Module II					
Natural rubber	Discuss the source of Natural rubber,Monomer,structure of isoprene,Polyisoprene,Molecular weight of NR.	L	CO2	Understand	1
Properties	List 5 properties of NR	L	CO2	Understand	1
Latex	Differentiate field latex ,concentrated latex. - centrifuged and double centrifuged latex.	L	CO2	Understand	1
Preservation	Explain the preservation of NR latex and concentration methods.-centrifuging and creaming.	L	CO2	Understand	1
Latex products	Familiarise various latex products available in the market and various job opportunities in latex industries.	L	CO2	Understand	1
Forms of NR	Introduce different forms of dry rubber-RSS ,ADS ,Crepe rubber and ISNR.	L	CO2	Understand	1
Manufacture	Outline the production of RSS,ADS	L	CO2	Understand	1
Crepe rubber,ISNR.	Outline the production of Crepe rubber and ISNR.	L	CO2	Understand	1
Dry rubber industries	Familiarise various industries manufacturing different marketable forms of NR in Kerala.	L	CO2	Understand	1
Guttapercha	Differentiate NR and Guttapercha. Structure of Cis NR and Trans NR	L	CO2	Understand	2
RSS,ISNR	Prepare samples of RSS and ISNR.	P	CO2	Apply	1.5
Preservation	Demonstrate the preservation of NR Latex.	P	CO2	Apply	2
Cogulation	Demonstrate the coagulation of NR Latex.	P	CO2	Apply	3
NR products	Prepare 6 samples of products made of NR with their property and application - Tyre,InnerTube,Latex gloves,Balloon,Hawai,latex foam,Sponge.	P	CO2	Apply	3
	Lab Series Exam - I	P	CO2	Apply	1.5
Module III					
Synthetic Rubber	Discuss the raw material for synthetic rubber and what is synthetic rubber.	L	CO3	Understand	1
GP,SP Rubber	Identify the names of General purpose and special purpose synthetic rubbers.	L	CO3	Understand	1

SBR,BR,IR	Explain the structure , 5 properties and applications of IR,SBR ,BR ,IIR.	L	CO3	Understand	3
EPDM,NBR,CR	Explain the structure ,5 properties and applications of EPDM,NBR,CR.	L	CO3	Understand	3
CSM,Silicone	Explain the structure, 5 properties and applications of CSM and Silicone.	L	CO3	Understand	3
SBR,BR	Collect samples of products made of SBR ,BR. and label its properties and applications.	P	CO3	Apply	3
IIR,EPDM	Collect samples of products made of IIR,EPDM. and label its properties and applications.	P	CO3	Apply	3
NBR,CR	Collect samples of Products made of NBR,CR. and label its properties and applications.	P	CO3	Apply	3
Silicone rubber	Collect samples of products made of Inorganic polymer.- silicone rubber. and label its properties and applications.	P	CO3	Apply	3
Module IV					
Classify plastics	Classify plastics and define thermoplastics and thermosets with 5 examples each,Advantages and disadvantages of plastics.	L	CO4	Understand	1
Kitchen ware plastics	Identify the polymer used to makeWater/milk bottles, food containers (Tupperware), sandwich bags, plastic wrap, cutlery, microwave-safe dishes, bottle caps, dustbins, buckets, straws.	L	CO4	Understand	1
bathroom used plastics	Identify the plastic material used to make Toothbrushes, cosmetic containers, razors, shower curtains, shampoo/soap bottles, wet wipes.	L	CO4	Understand	1
cleaning and storage	Identify the plastic used to make Spray bottles for cleaners, trash bags, storage bins, hoses, pipes.	L	CO4	Understand	1
Electronic goods	Identify the plastic used to make Mobile phone covers, toy components, pens, notebook covers, packaging materials for online orders, and internal components for electronics.	L	CO4	Understand	1
Furniture	Identify the plastic material used to make Plastic chairs, tables, storage crates, insulation sheets.	L	CO4	Understand	1
Fibres	Define fibre and Classify fibres	L	CO4	Understand	1
Fibres	Explain natural and synthetic fibres with five examples each.	L	CO4	Understand	1
Natural and synthetic fibres	List the Properties and applications of wool, coir, silk, Arecanut, Nylon, acrylic, PE, PP, Carbon, glassfibres.	L	CO4	Understand	1
other polymer forms	Explain Liquid resins,adhesives and sealants with 2 examples each.	L	CO4	Remember	1
Raw Plastics	Collect 3 samples of Thermoplastics and three samples of thermoset polymer and label name of polymer.	P	CO4	Apply	1.5
Plastic products	Collect 3 samples of products made of LDPE,HDPE and PVC and label the name of polymer,name ,3 properties and its application.	P	CO4	Apply	3
Natural Fibre	Collect 3 samples of natural fibres and label its name ,3 properties and its application.	P	CO4	Apply	3
Synthetic fibre	Collect 3 samples of Synthetic fibre,label its name ,3 properties and its application.	P	CO4	Apply	2
	Lab Series Exam - II		CO4	Remember	1.5

Suggested Student Activities for Self-Learning

Week	Self-Learning description	Hours
Module I		
1	Explain the relationship between molecular structure (monomers, repeating units,) and material properties like strength, flexibility, and transparency. Collect 2 dried samples of natural polymer and search from the internet and learn their 3 main properties and its application.	6
2	Illustrate the conceptual representation of polymer molecules by linking together simple molecules. Collect 2 dried samples of synthetic polymer and learn its 3 main properties and applications by searching from the internet .	6

3	Explore what is Bakelite. Collect 5 samples of organic polymer ,and learn its 3 main properties and its applications by searching from the internet.	6
4	Find 10 monomers and list the polymers made from them. Collect a sample of inorganic polymer and learn its 3 main properties and applications from the internet.	6
Module II		
5	Explore different trees from which latex can be taken. Collect samples of RSS and ISNR and learn the difference of properties between them.	6
6	Compare creaming and centrifuging. Search from the internet the different preservatives used to preserve NR latex .	6
7	Find out the different centrifuging units in kerala. Make a visit to the latex industry and dry rubber industry. Search from the internet the chemical reaction happening while coagulating NR latex.	6
8	Learn the chemical name and structure of Natural rubber. Understand the difference between a latex product and a dry rubber product.	6
Module III		
9	Search from the internet what are the different products made of synthetic rubbers? Collect samples of products made of SBR and BR and study its 3 properties and 3 applications.	6
10	Learn the structure of CSM and Silicone rubber. Collect samples of products made of IR and EPDM and study its 3 properties and 3 applications.	6
11	Learn the structure of SBR ,IR and BR. Collect samples of products made of NBR and CR and study its 3 properties and 3 applications.	6
12	Learn the structure of EPDM,NBR and CR. Collect samples of product made of CSM and silicone and study its 3 properties and 3 applications.	6
Module IV		
13	Search the internet and find 10 different products made of thermoplastics and thermosets. Collect 3 samples of raw thermoplastics and 3 samples of thermosets and study their name,properties and applications.	6
14	Search the internet and learn about the properties required for fibre forming polymers. Collect 3 samples of natural fibres and study their name,properties and applications.	6
15	Search the internet and learn about the application of fibres . Collect three samples of synthetic fibres and study their name, properties and applications.	6
Total Self-Learning hours		90

Learning Resources

TEXT BOOK			
Sl. No.	Title	Author	Publication
1	Polymer science	Jayadev Sreedhar, V R Gowariker, N.V Viswanathan	New Age International (P) Ltd, New Delhi- 2005 Edition
2	Text Book of Polymer Science.	Billmeyer.J.R	(John Wiley & Sons (Aia) PTE Ltd, Singapore, 2008 Edition
3	Man made Fibres	R.W.Moncrieff	Wiley 5th edition.1970
4	Hand book of natural rubber production -Rubber board	RRII,Kottayam	

REFERENCE			
Sl. No.	Title	Author	Publication
1	Introductory Polymer Chemistry.	G.S .Misra	New Age International (P) Ltd, New Delhi- 2005 Edition
2	Introduction to Polymerscience and Technology	Mustafa Akay	bookboon.com

WEB RESOURCES		
Sl. No.	URL	Modules Covered
1	https://study.com	I
2	https://www.osborneindustries.com	III
3	https://www.sealsdirect.co.uk	II

4	https://www.sciencedirect.com	All
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Major Equipment/Instruments/Components required for a batch of 30 students

Sl. No.	Particulars	Specifications	Quantity
	NIL		

Major Software required for a batch of 30 students

Sl. No.	Particulars	Specifications	Quantity
	NIL		

Assessment Methodology - Practicum

Mode	Attendance	CA1	CE	CA2	CA3	CA4	CA5	ESE	
		Average of Self learning activities from each module	Continuous evaluation	Practical test (first half of experiments)	Practical test (second half of experiments)	Series Exam (2 modules)	Series Exam (2 modules)	Written Exam (theory)	Lab Examination (internal)
Duration				3 hours	3 hours	2 hours	2 hours	2.5 hours	3 hours
Exam mark		10	50	40	40	50	50	60	40
Converted to			10	10	10	10	10		
Marks	5	5	10	10		10		60	40
Total								100	
Tentative schedule	End of semester	By 4th week	By 7th week	By 11th week	By 14th week	8th week	15th week	End of semester	

Scheme of Evaluation - Practical (Continuous Internal Evaluation)

Sl. No.	Criteria	Description	Marks
1	Preparation & Punctuality	Attendance, timely submission of record, pre-lab preparation, understanding of experiment objectives and theory.	10
2	Experimental Setup & Procedure	Ability to identify components/equipment, correct circuit connections/setup, systematic and safe execution of the procedure.	10
3	Observation & Data Recording	Accuracy and neatness of readings, correct use of units, tabulation, and systematic recording of results.	5
4	Analysis & Result Interpretation	Proper calculations, graphical representation, result verification, discussion, and logical conclusion.	10
5	Viva Voce / Concept Understanding	Clarity of theoretical concepts, explanation of procedure, ability to answer questions related to the experiment.	10
6	Workmanship, Discipline & Teamwork	Laboratory discipline, care of instruments, teamwork, attitude, and safety practices.	5
Total			50

Scheme of Evaluation - Practical Test

Category	Things to evaluate	Marks
Write-up / Procedure (before experiment)	Aim, circuit diagram/flowchart, stepwise procedure.	10
Experiment Setup & Execution	Correct circuit connections / coding / setup. Handling of instruments/software. Accuracy of execution.	10
Observation & Result / Output	Proper tabulation of readings. Calculations/graphs/program output. Correctness of result / expected outcome.	8
Viva Voce	Conceptual understanding of experiment. Related theoretical knowledge.	8
Record	Completion & neatness of practical record. Proper certification by faculty.	4
Total		40

Series Examination Pattern- Theory

Time	Module	Part A	Marks	Part B	Marks	Part C (With choice)	Marks	Total
2 hours	Any one module	2	1	3	3	2	7	25
	Any one module	2	1	3	3	2	7	25
	Number	4		6		4		50

End Semester Examination Pattern of Question Paper - Theory

Duration	Module	TYPE OF QUESTIONS						
		Part A (2 Questions from each module)		Part B 2 out of 3 questions from each module)		Part C (1 set questions with choice from each module)		Total Marks
		No. of Questions	Marks (1 mark each)	No. of Questions	Marks (3 marks each)	No. of Questions	Marks (7 marks each)	
2.5 hours	1	2	2	3	6	1 set	7	15
	2	2	2	3	6	1 set	7	15
	3	2	2	3	6	1 set	7	15
	4	2	2	3	6	1 set	7	15
		8	8	12	24	4 sets	28	60

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